

talk to various TTS engines, including eSpeak, Flite and Festival. The last one has the best voices but these are easily available only for English.

You will need to ensure that several software packages are installed. The minimal list for Fedora 10 and Ubuntu 8.10 is:

- *espeak*
- *espeak-data* also on Ubuntu
- *speech-dispatcher*
- *speech-dispatcher-python* (on Fedora)
- *python-speechd* (on Ubuntu)

You may wish to enter some text in Hindi. Indic Onscreen Keyboard (available on Fedora, but not on Ubuntu) is a reasonable option. The on-screen keyboard and the keyboard layouts are provided by *iok* and *m17n-contrib-hindi*.

You will need to configure the default engine in */etc/speech-dispatcher/speechd.conf* as follows:

```
DefaultModule espeak
```

Since Ubuntu and Fedora are now using PulseAudio, you should change the audio output in */etc/speech-dispatcher/modules/espeak.conf* to Pulse; otherwise, you may 'hear' only silence (on Fedora):

```
EspeakAudioOutputMethod "pulse"
```

You will need to create the default directory for the logs. The speech dispatcher will work but no log messages will be stored.

```
$ sudo mkdir /var/log/speech-dispatcher
```

Now, start the *speech-dispatcher* service and you are ready to start.

```
$ sudo service speech-dispatcher start
```

Verify that the speech dispatcher is working:

```
$ spd-say "Hello. Hello. Testing 1 2 3"
```

If you hear what you expect, you can now proceed further.

Learning the first steps

You need to learn what the speech dispatcher provides and how it behaves. Interactive learning is the easiest. So, start your Python interpreter and try the following:

```
>>> import speechd
>>> dir(speechd)
>>> help(speechd.Speaker)
```

A little exploring and you realise that you need to

create an object of the type *Speaker*. So, create it and see what happens next:

```
>>> spk = speechd.Speaker('me')
>>> dir(spk)
```

List commands are always helpful, so try the following:

```
>>> spk.list_output_modules()
('espeak', 'flite')
```

The first command tells you the output TTS engines available. If the default is not eSpeak or you wish to switch between the modules, you can easily do so:

```
>>> spk.set_output_module('espeak')
>>> spk.speak('Testing 1 2 3')
(225, 'OK MESSAGE QUEUED', ('3',))
```

That was easy, right? Now, list the voices available:

```
>>> spk.list_synthesis_voices()
(('afrikaans', 'af', 'none'), ... ('hindi-test', 'hi', 'none'), ... ('cantonese-test', 'zh', 'yue'))
>>> spk.set_language('hi')
```

Hindi is an option, though only as a testing version, at present. Try to say something:

```
>>> spk.speak('Testing 1 2 3')
(225, 'OK MESSAGE QUEUED', ('4',))
```

You should have noticed a difference. Maybe it was too fast so let us slow it down:

```
>>> help(spk.set_rate)
>>> spk.set_rate(-50)
>>> spk.speak('Testing 1 2 3')
```

Ascribing negative values while setting the rate, slows down the speech. The numbers should clearly be spoken in Hindi.

```
>>> help(spk.set_voice)
>>> spk.set_voice('FEMALE1')
>>> spk.speak('Testing 1 2 3')
>>> spk.set_pitch(50)
>>> spk.speak('Testing 1 2 3')
```

Continue your exploring. The voice isn't very feminine, so there's more potential for work!

What if you enter a Hindi text message? Try using the on-screen keyboard, *iok* and enter a Unicode message:

```
>>> text=u"नमो सिंह कहते थे, जो डर गया वह नर गया."
>>> spk.speak(text)
```